IDBC protobuff V0.2

```
syntax = "proto3";
package IBDC;

// Embedded system sends this at event detection
message EventNotification {
  uint32 event_id = 1; // unique ID for each event
  uint32 distance_cm = 2; // Distance to detected object in cm
  uint32 time_offset_ms = 3; Time between detection and notification in ms
  uint32 image_count = 4; Number of images associated w/ this event
  ImageFormat image_format = 5; Image format used for this event.
}

// support different image formats
enum ImageFormat {
  JPEG = 0;
  PNG = 1;
}

// metadata describing an individual image.
// Phone can use this information to validate and track image transfer.
message ImageInfo {
  uint32 event_id = 1; // Event this image belongs to
  uint32 image_index = 2; // Image number within the event
  uint32 image_size_bytes = 3; // Total image size in bytes
  uint32 total_chunks = 4; // Total number of BLE chunks for this image
  ImageFormat image_format = 5; // format of this image
}

// Header and payload for each image chunk sent over BLE.
message ImageChunk {
  uint32 event_id = 1; // Event this image belongs to.
  uint32 image_index = 2; // which image this chunk belongs to.
  uint32 chunk_sequence = 3; // Zero-based chunck sequence #.
  uint32 total_chunks = 4; // Total number of chunks for this image
  bool is_last_chunck = 5; // True if this is final chunk.
  bytes payload = 6; // Raw image bytes contained in this chunck.
}
```

```
// phone request an image transfer.
// can be used for initial transfer, delayed transfer, resuming after
// a disconnect
message ImageTransferRequest
  uint32 event_id = 1; // Event whose images are requested.
  uint32 start_from_image = 2; // zero-based image index to start from
  uint32 start_from_chunk = 3; // zero-based chunk to start from
  }

// phone ack successful receipt of an entire Event
// Note: will IBDC delete events to save space? this will be useful if yes.
 Message EventTransferAck {
   uint32 event_id = 1; // Event successfully received by phone.
  }

// Phone writes settings to embedded system
message Settings {
   uint32 images_per_event = 1; // # of images captured per event
    }

// Embedded system exposes current device status.
message DeviceStatus {
  uint32 protocol_version = 1; // protocol version used by device
  uint32 battery_percent = 2; // Battery percent 0-100
  uint32 pending_events = 3; // # of events waiting to be transfered
  uint32 storage_available_percent = 4; // available storage 0-100%

// Phone request a list of pending events
  message PendingEventListRequest {
    repeated uint32 event_ids = }
```

// Empty message
// because list requires
// no parameters

```
// Embedded system responds with IDs of events that
// have not been acknowledged by the phone
message PendingEventList {
  repeated uint32 event_ids = 1;
}

// Phone request information for a specific event
// eg. original EventNotification was missed, phone reconnects later,
// phone discovers an event through PendingEventList
message EventInfoRequest {
  uint event_id = 1;
    }
```